CLASS 8 · ICSE · SELINA CONCISE MATHEMATICS · SHORT NOTES

Chapter 7: Percent and Percentage

Topic-wise short notes with solved examples

The meaning of per cent, converting between percentages, fractions, decimals and ratios, finding a percentage of a quantity, percentage increase and decrease, and word problems.

1 Per cent

- **Per cent** means "per hundred" or "out of a hundred". The symbol is %.
- $x\% = \frac{x}{100}$. For example $35\% = \frac{35}{100} = \frac{7}{20} = 0.35$.

To convert	Do this	Example
Per cent to fraction	Divide by 100 and simplify	$45\% = \frac{9}{20}$
Fraction to per cent	Multiply by 100	$\frac{3}{8} = 37.5\%$
Per cent to decimal	Move the decimal point two places left	$6.5\% = 0.065$
Decimal to per cent	Move the decimal point two places right	$0.48 = 48\%$
Per cent to ratio	Write over 100 and simplify	$25\% = 1 : 4$
Ratio to per cent	Write as a fraction and multiply by 100	$3 : 5 = 60\%$

- Useful values: $50\% = \frac{1}{2}$, $25\% = \frac{1}{4}$, $20\% = \frac{1}{5}$, $12\frac{1}{2}\% = \frac{1}{8}$, $33\frac{1}{3}\% = \frac{1}{3}$, $66\frac{2}{3}\% = \frac{2}{3}$.

Example 1.1 Express $\frac{7}{25}$, 0.085 and $2 : 5$ as percentages.

SOLUTION

28%, 8.5% and 40%

Example 1.2 Express $16\frac{2}{3}\%$ as a fraction.

SOLUTION

$\frac{50}{3} \times \frac{1}{100}$

$\frac{1}{6}$

2 Percentage of a quantity

- $x\%$ of a quantity $= \frac{x}{100} \times$ the quantity.
- To express one quantity as a percentage of another (same units): $\frac{\text{first}}{\text{second}} \times 100\%$.

- To find the whole when a percentage of it is known: whole $=$ part $\times \dfrac{100}{\text{percentage}}$.

Example 2.1 Find 15% of ₹840 and $12\frac{1}{2}\%$ of 64 kg.

SOLUTION

$\dfrac{15}{100} \times 840$ and $\dfrac{1}{8} \times 64$

₹126 and 8 kg

Example 2.2 What per cent of 2 hours is 45 minutes?

SOLUTION

$\dfrac{45}{120} \times 100$

37.5%

Example 2.3 30% of a number is 72. Find the number.

SOLUTION

$72 \times \dfrac{100}{30}$

240

Example 2.4 A number exceeds 20% of itself by 40. Find the number.

SOLUTION

$x - 0.2x = 40$, so $0.8x = 40$

50

3 Increase and decrease

$$\text{Percentage change} = \frac{\text{change}}{\text{original value}} \times 100\%$$

- After an **increase** of $x\%$, new value $=$ original $\times \left(1 + \dfrac{x}{100}\right)$.
- After a **decrease** of $x\%$, new value $=$ original $\times \left(1 - \dfrac{x}{100}\right)$.
- To find the **original** value: divide the new value by the same multiplier.

Example 3.1 The price of sugar rises from ₹40 to ₹46 per kg. Find the percentage increase.

SOLUTION

$\dfrac{6}{40} \times 100$

15%

Example 3.2 A salary of ₹24,000 is increased by 12.5%. Find the new salary.

SOLUTION

$24000 \times \frac{9}{8}$

₹27,000

Example 3.3 After a 20% decrease, the price of a book is ₹160. Find the original price.

SOLUTION

$160 \div 0.8$

₹200

Example 3.4 A number is increased by 20% and the result is then decreased by 20%. Find the net change.

SOLUTION

$1.2 \times 0.8 = 0.96$

A decrease of 4%

4 Comparisons

- If A is $x\%$ **more** than B, then B is $\frac{x}{100+x} \times 100\%$ **less** than A.
- If A is $x\%$ **less** than B, then B is $\frac{x}{100-x} \times 100\%$ **more** than A.
- If a price rises by $x\%$, consumption must fall by $\frac{x}{100+x} \times 100\%$ to keep the expenditure unchanged.

Example 4.1 A's income is 25% more than B's. By what per cent is B's income less than A's?

SOLUTION

$\frac{25}{125} \times 100$

20%

Example 4.2 The price of rice rises by 25%. By what per cent must a family reduce its consumption so that its expenditure on rice does not change?

SOLUTION

20%

5 Word problems

Example 5.1 In an examination a student scored 432 marks out of 600. Find the percentage. If the pass mark is 40%, how many marks are needed to pass?

SOLUTION

$\frac{432}{600} \times 100 = 72$ and $0.4 \times 600 = 240$

72%; 240 marks

Example 5.2 A candidate scores 35% and fails by 30 marks. Another scores 42% and gets 12 marks more than the pass mark. Find the maximum marks and the pass mark.

SOLUTION

$0.35M + 30 = 0.42M - 12$, so $0.07M = 42$ and $M = 600$.

Maximum marks 600; pass mark 240

Example 5.3 In an election between two candidates, the winner got 58% of the votes and won by 2400 votes. Find the total number of votes.

SOLUTION

The loser got 42%. The difference is 16% of the votes: $0.16T = 2400$.

15,000

Example 5.4 An alloy contains 36% zinc, 40% copper and the rest nickel. Find the mass of nickel in 500 g of the alloy.

SOLUTION

Nickel is 24%: 0.24×500

120 g

Example 5.5 A man spends 75% of his income and saves ₹4500. Find his income.

SOLUTION

Savings are 25% of the income.

₹18,000

Example 5.6 The population of a town increases by 10% every year. If it is 50,000 now, find it after 2 years.

SOLUTION

$50000 \times 1.1 \times 1.1$

60,500

6 Common mistakes

- Using the new value as the base for a percentage change. The base is the original value.
- Thinking a 20% rise followed by a 20% fall leaves the value unchanged.
- Comparing quantities in different units (minutes with hours) without converting.
- Finding the original value by subtracting the percentage from the new value.
- Writing 0.5% as 0.5 instead of 0.005.
- Treating "A is 25% more than B" as "B is 25% less than A".

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